

Question 1: System Objective and Deterministic Baseline

Project Objective

The objective of the project is to maximize the probability of arriving at a destination before a calculated deadline by selecting routes that account for traffic congestion effects. This simulation model studies how route planning performs when travel times are affected by real-world traffic uncertainty such as congestion, incidents, and demand fluctuations.

This project mainly evaluates how reliable deterministic routing strategies are when travel times vary stochastically across simulation runs. The model uses traffic simulation to model uncertainty and measures the probability of reaching the destination within the desired time.

Deterministic Baseline Strategy

The deterministic baseline implemented in the project is fixed route selection using the A* shortest path algorithm. In this approach, the A* (A-star) algorithm computes a route that minimizes the expected total travel time based on deterministic edge weights. Once the route is selected, the route remains fixed throughout the simulation and the travel times may vary between simulation runs, but the routing decision does not change.

Implementation

In the implementation:

- The road network is parsed from the file `map.net.xml`.
- A start edge and destination edge are selected.
- The A* algorithm computes the shortest route between them.
- The algorithm evaluates candidate edges using the cost function:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = cost from the start node to node n (accumulated edge length)
- $h(n)$ = heuristic estimate to the destination (Euclidean distance)
- In the provided code:
 - g is accumulated using edge lengths (`next_edge.getLength()`).

- h is computed using the Euclidean distance between nodes.
- The route returned by A^* is then used as a fixed deterministic path for the vehicle in the SUMO simulation.

Reason for Choosing this Baseline

We have chosen this deterministic routing baseline because:

1. A^* algorithm is a standard shortest-path (deterministic) algorithm which is computationally efficient and suitable for real-time navigation systems. Highly efficient applications such as Google Maps use this algorithm (in combination with other algorithms) to give the shortest path.
2. It efficiently combines the actual path cost (time taken) and a heuristic estimate to the destination.
3. The algorithm guarantees optimal shortest paths when the heuristic is admissible.
4. The routing decision remains deterministic, meaning the same inputs produce the same route.
5. It provides a clear baseline against which other probabilistic or risk-aware routing strategies can later be compared.

This aligns with our modeling assumption that routing decisions are deterministic even though the environment may be uncertain. Therefore, the deterministic A^* routing strategy is considered as a standard baseline for evaluating routing reliability under uncertainty as a part of this project.

Question 2: Random Variables in the Simulation

Although the routing decision is deterministic, the environment is considered stochastic because traffic conditions cause travel times to vary. The key random variables used in the system are described below.

1. Segment Travel Time

Definition

Let T_i be the travel time of the i -th road segment of the selected route. Each road segment has a travel time that varies depending upon the traffic conditions.

Distribution

In the simulation, each segment travel time is modeled as

$$T_i \sim \text{Lognormal}(\mu_i, \sigma_i^2)$$

We have chosen the lognormal distribution because the travel times (T_i) are always positive. Moreover, traffic delays generally create right-skewed distributions because delays only increase the travel time. The lognormal distribution also closely matches observed traffic delay patterns.

Role in Simulation

In the simulation:

1. The deterministic A* algorithm selects a fixed route.
2. For each simulation iteration, a random value of T_i is sampled for each segment.
3. The total travel time for the route is calculated.

Hence, even though the route is fixed, the travel time experienced on each segment changes between runs.

2. Total Route Travel Time

Definition

Let T_{route} represent the total travel time for the entire route. It is computed as the sum of the travel times across all road segments:

$$T_{\text{route}} = \sum_{i=1}^n T_i$$

where n is the number of segments in the route.

Distribution

Since each T_i is lognormally distributed, the total route time becomes the sum of lognormal variables, giving a stochastic distribution for the overall travel time.

Role in the Simulation

For each simulation run k :

$$T_{\text{route}}^{(k)} = \sum_{i=1}^n T_i^{(k)}$$

The total route time is then used to evaluate performance relative to the deadline.

3. Arrival Deviation

Definition

Arrival deviation is defined as:

$$A = T_{\text{route}} - D$$

where T_{route} is the total route travel time and D is the deadline for arrival.

Interpretation

- $A < 0 \rightarrow$ early arrival
- $A = 0 \rightarrow$ on-time arrival
- $A > 0 \rightarrow$ late arrival

Role in the Simulation

This random variable measures the reliability of the chosen route. By analyzing the distribution of A , the system estimates

$$P(T_{\text{route}} < D)$$

which represents the probability of arriving before the deadline.

End of Submission